

Nithin Bobbiligama

Senior Data Engineer
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PROFESSIONAL SUMMARY

Senior Data Engineer with 6+ years of experience architecting and delivering large-scale data platforms across banking, financial services, insurance, pharmaceutical, and fintech domains. Deep expertise in cloud-native lakehouse design, real-time streaming pipelines, and medallion-based ETL/ELT architecture on AWS and Azure. Proven track record leading mission-critical pipelines for AUM reporting, credit risk analytics, fraud detection, and clinical data governance. Proficient in Python, Apache Spark, Kafka, Snowflake, Delta Lake, and infrastructure-as-code with Terraform. Experienced in PCI-DSS, GDPR, FCRA, HIPAA, and GxP compliance frameworks. Strong technical leadership in design reviews, cross-functional collaboration, and mentoring data engineering teams.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, SQL, PySpark, SAS, Bash/SSH
- **Cloud Platforms:** Azure (Data Factory, Databricks, Synapse, ADLS Gen2, Event Hub, Functions, Logic Apps), AWS (S3, Glue, EMR, Redshift, Lambda, Step Functions, DMS, CloudWatch, RDS)
- **Big Data & Streaming:** Apache Spark, PySpark, Apache Kafka, Apache Airflow, Delta Lake, DBT, Apache Beam, Flink, Hive, HBase, MapReduce, Oozie, IBM DataStage
- **Databases & Warehousing:** Snowflake, Azure Synapse, Redshift, PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Cosmos DB, Teradata, DB2
- **Data Governance & Quality:** Great Expectations, custom quality frameworks, schema validation, audit logging, data lineage, CDISC/SDTM, PCI-DSS, FCRA, GDPR, HIPAA, GxP
- **CI/CD & DevOps:** Terraform, Docker, GitHub Actions, Jenkins, Azure DevOps, Git (GitHub, GitLab, Bitbucket)
- **Visualization:** Power BI, Tableau, Amazon QuickSight
- **Methodologies:** Agile, Scrum, Kanban, SDLC

WORK EXPERIENCE

- **Nationwide** Nov 2024 – Present
Senior Data Engineer The colony TX
 - **AUM & NNC Reporting:** Architected and owned month-end AUM and NNC reporting pipelines using Azure Data Factory and Apache Airflow, delivering accurate financial aggregates to investment and wealth management leadership under strict SLA windows.
 - **Lakehouse Migration:** Led migration of legacy ETL workloads to Azure Databricks with Apache Spark, refactoring high-volume transformations and cutting month-end AUM calculation processing time by 40%, directly accelerating financial close cycles.
 - **Real-Time Fraud Detection:** Designed and deployed a real-time fraud detection pipeline using Azure Event Hub and Kafka, streaming high-risk claims and suspicious transactions to security operations for immediate regulatory response.
 - **Delta Lake & Audit:** Implemented Delta Lake on ADLS Gen2 with full ACID transaction support and time-travel capabilities, enabling reliable audit trails and point-in-time recovery for month-end reconciliation and regulatory examination.
 - **Metadata-Driven Ingestion:** Engineered a metadata-driven ingestion framework in Python that standardized source onboarding across policy, claims, and financial systems, reducing data latency from weeks to days and eliminating manual onboarding effort.
 - **Compliance Automation:** Automated NAIC compliance reporting and month-end NAV calculations using Python and Azure Functions, replacing error-prone Excel workflows and accelerating financial close by two business days per cycle.
 - **Medallion Architecture:** Designed bronze/silver/gold medallion architecture on ADLS Gen2 with Great Expectations data quality gates, ensuring high-confidence datasets for downstream analytics and executive dashboards.
 - **Infrastructure as Code:** Managed complete infrastructure lifecycle using Terraform across ADLS Gen2, Azure Data Factory, and Databricks workspaces spanning dev, test, and production environments.
 - **Snowflake Optimization:** Optimized Snowflake query performance using clustering keys, materialized views, and query profiling, reducing dashboard refresh times by 35% for business stakeholders consuming month-end financial reports.
- **Comcast** Jul 2023 – Nov 2024
Data Engineer Dayton, OH
 - **Streaming Data Platform:** Designed and operated high-throughput Kafka streaming pipelines ingesting subscriber activity, viewership telemetry, and billing events across millions of endpoints, enabling near-real-time analytics for product and revenue teams.
 - **Data Lakehouse Architecture:** Built and maintained a lakehouse on AWS S3 using Delta Lake and Apache Spark, structuring bronze/silver/gold layers for subscriber behavior, content performance, and churn prediction datasets.
 - **Pipeline Orchestration:** Engineered Airflow DAGs and AWS Step Functions workflows for daily and hourly data refresh cycles, incorporating SLA monitoring, dependency management, and automated retry logic for critical pipelines.
 - **Snowflake Warehousing:** Led migration of analytical workloads to Snowflake, implementing role-based access control, dynamic data masking for PII compliance, and multi-cluster warehousing to support concurrent BI and data science query loads.
 - **Cost Optimization:** Reduced AWS EMR and Glue processing costs by 30% through Spark job tuning, partition pruning, and intelligent data tiering strategies across hot and cold S3 storage layers.
 - **Data Quality Framework:** Implemented automated data quality checks using Great Expectations and custom Python validators on ingestion, transformation, and serving layers, reducing incident escalations from downstream consumers by 40%.

- **Cross-Functional Leadership:** Partnered with data science, product analytics, and BI teams to define data contracts and schema standards, driving adoption of a unified data catalog and reducing ad-hoc engineering requests by 25%.
- Experian** Jul 2020 – Jul 2021
Data Engineer Hyderabad, India
- **Credit Analytics Pipelines:** Designed end-to-end data solutions for banking and credit analytics teams using Python, SQL, and Spark on AWS (S3, Glue, EMR, Redshift), translating complex business requirements into scalable architectures for credit scoring and risk analytics.
 - **Pipeline Orchestration:** Orchestrated daily credit data refresh and underwriting pipelines using Apache Airflow and AWS Step Functions, ensuring reliable scheduling, SLA adherence, and automated retries for critical scoring workflows.
 - **PII & Compliance:** Masked sensitive PII and payment card data using dynamic data masking, hashing, and field-level redaction techniques, maintaining full PCI-DSS and FCRA compliance across non-production environments.
 - **Data Transformation:** Transformed nested JSON and XML payloads into query-optimized columnar structures using PySpark and AWS Glue, enabling faster analytical queries and accelerating BI reporting turnaround.
 - **Real-Time Fraud Monitoring:** Built real-time fraud detection and credit monitoring pipelines using Kafka and AWS Kinesis, reducing event processing latency by 50% during peak transaction periods.
 - **Warehouse Optimization:** Optimized credit scoring workloads in Snowflake and Redshift using clustering, partitioning, and materialized views, standardizing data lake storage with Parquet and Delta formats on S3.
 - **Mentorship & CI/CD:** Led technical design reviews, mentored junior engineers on data governance best practices, and streamlined deployment pipelines using Git, Jenkins, and GitHub Actions.
- Novartis** Jan 2019 – Jul 2020
Data Engineer Hyderabad, India
- **Clinical Data Pipelines:** Structured clinical trial and lab data requirements with research stakeholders, translating business needs into technical ingestion pipeline specifications using APIs and batch ETL on AWS (S3, Glue, EMR).
 - **Regulatory Compliance:** Ensured CDISC and SDTM model compliance by validating clinical and patient datasets with SQL transformations, supporting successful regulatory submissions to global health authorities.
 - **Pharmacovigilance Streaming:** Monitored pharmacovigilance and adverse event data flows using Kafka streaming, analyzing event latency and coordinating with safety teams to improve alert throughput for critical safety signals.
 - **GxP Pipeline Orchestration:** Orchestrated GxP-validated data pipelines using SAS and Airflow, managing scheduling, error handling, and audit logging for daily clinical research workflows.
 - **Genomic Data Processing:** Partnered with genomic and real-world evidence research teams to optimize Spark transformations, cutting processing time for terabyte-scale clinical datasets while documenting full lineage for reproducibility.
 - **Data Quality:** Implemented data quality frameworks using Python and SQL validations, reducing discrepancies in clinical trial datasets and generating compliance quality reports for study monitors and CRO partners.
 - **Automation:** Automated manual data reconciliation between CROs and internal systems using Python and SAS scripting, reducing reconciliation cycle time by 60% and eliminating error-prone manual steps.

EDUCATION

- **University of Dayton** Aug 2021 – May 2023
Master of Science – Computer Science Dayton, OH
 - **Relevant Coursework:** Advanced Database Systems, Distributed Computing, Cloud Architecture, Machine Learning, Big Data Analytics
- **Mahaveer Institute of Science and Technology** Aug 2015 – May 2019
Bachelor of Technology – Computer Science & Engineering India
 - **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

TECHNICAL PROJECTS

- **Real-Time Fraud Detection Engine:** Architected an end-to-end streaming fraud detection system using Kafka, PySpark Structured Streaming, and Delta Lake. Implemented ML feature pipelines with sliding-window aggregations and anomaly scoring, reducing false positive rates by 22% compared to batch-based baselines.
- **Medallion Lakehouse Platform:** Designed a production-grade bronze/silver/gold lakehouse on ADLS Gen2 with automated Great Expectations quality gates, Delta Lake ACID guarantees, and Terraform-managed infrastructure, supporting 15+ downstream analytics consumers.
- **Metadata-Driven Ingestion Framework:** Built a configuration-driven Python ingestion framework capable of onboarding new data sources via JSON manifest files, eliminating bespoke pipeline code and reducing source onboarding time from 2 weeks to under 1 day.
- **Clinical Trial Data Quality System:** Developed a GxP-compliant data quality and lineage tracking system for clinical trial datasets using Python, SQL, and Airflow, producing audit-ready quality reports consumed by study monitors and regulatory submission teams.

CERTIFICATIONS

- **Microsoft:** Microsoft Certified Azure Data Engineer Associate
- **Microsoft:** Microsoft Certified Power BI Data Analyst Associate
- **AWS:** AWS Certified Data Engineer – Associate
- **AWS:** AWS Certified Solutions Architect – Associate
- **Databricks:** Databricks Certified Associate Developer for Apache Spark
- **Snowflake:** Snowflake SnowPro Core Certification